

## PRACTICAL COURSE WORKBOOK

# Python for Finance & Quant Analysis

Build a clearer, auditable financial workflow

|             |                                                                |
|-------------|----------------------------------------------------------------|
| LEVEL       | Intermediate                                                   |
| GUIDED TIME | 5 hours across 12 lessons                                      |
| PROJECT     | a fictional financial model with assumptions and checks        |
| SKILLS      | Python, yfinance, NumPy, Portfolio, Quant, Critical evaluation |
| EDITION     | Structured draft - version 0.9                                 |

## WHAT YOU WILL BE ABLE TO DO

- Use Python appropriately in a realistic, bounded task.
- Use yfinance appropriately in a realistic, bounded task.
- Use NumPy appropriately in a realistic, bounded task.
- Use Portfolio appropriately in a realistic, bounded task.

## WHO THIS IS FOR

Learners studying financial technology and analytical methods.

## BEFORE YOU BEGIN

- Basic percentages and spreadsheet skills

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Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

## WORKBOOK GUIDE

# A clear route through the course

Python for Finance & Quant Analysis is a structured, practical course covering Python, yfinance, NumPy, Portfolio, Quant. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

| MODULE                                   | LESSONS AND FOCUS                                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------|
| <b>01 Python: Financial foundations</b>  | 1. Value with Python<br>2. Risk with yfinance<br>3. Markets with NumPy                |
| <b>02 yfinance: Tools and data</b>       | 1. Records with Portfolio<br>2. Analysis with Quant<br>3. Controls with Python        |
| <b>03 NumPy: Applied decisions</b>       | 1. Scenarios with yfinance<br>2. Evaluation with NumPy<br>3. Reporting with Portfolio |
| <b>04 Portfolio: Risk and governance</b> | 1. Security with Quant<br>2. Regulation with Python<br>3. Review with yfinance        |

## HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

## PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

## MODULE 01

# Python: Financial foundations

Apply Python through value, risk, markets.

## LESSON 01 / 18 MIN

## Value with Python

### Learning objectives

- Explain value in the context of Python for Finance & Quant Analysis.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

value, Python, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a value result produced with Python?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 02 / 22 MIN

## Risk with yfinance

### Learning objectives

- Explain risk in the context of Python for Finance & Quant Analysis.
- Apply yfinance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

risk, yfinance, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a risk result produced with yfinance?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 03 / 26 MIN

## Markets with NumPy

### Learning objectives

- Explain markets in the context of Python for Finance & Quant Analysis.
- Apply NumPy to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

markets, NumPy, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a markets result produced with NumPy?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

## MODULE REVIEW

# Turn the module into usable evidence

**1. RETRIEVE**

Without checking the lessons, explain the module's three most important ideas in your own words.

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**2. APPLY**

Describe the example or file you produced, the input used, and the result you observed.

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**3. REVIEW**

Record one uncertainty, one source to recheck, and the next deliberate practice action.

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## MODULE 02

# yfinance: Tools and data

Apply yfinance through records, analysis, controls.

## LESSON 04 / 30 MIN

## Records with Portfolio

### Learning objectives

- Explain records in the context of Python for Finance & Quant Analysis.
- Apply Portfolio to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

records, Portfolio, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a records result produced with Portfolio?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 05 / 18 MIN

## Analysis with Quant

### Learning objectives

- Explain analysis in the context of Python for Finance & Quant Analysis.
- Apply Quant to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

analysis, Quant, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a analysis result produced with Quant?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 06 / 22 MIN

## Controls with Python

### Learning objectives

- Explain controls in the context of Python for Finance & Quant Analysis.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

controls, Python, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a controls result produced with Python?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

## MODULE REVIEW

# Turn the module into usable evidence

**1. RETRIEVE**

Without checking the lessons, explain the module's three most important ideas in your own words.

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## MODULE 03

# NumPy: Applied decisions

Apply NumPy through scenarios, evaluation, reporting.

## LESSON 07 / 26 MIN

## Scenarios with yfinance

### Learning objectives

- Explain scenarios in the context of Python for Finance & Quant Analysis.
- Apply yfinance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

scenarios, yfinance, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a scenarios result produced with yfinance?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 08 / 30 MIN

## Evaluation with NumPy

### Learning objectives

- Explain evaluation in the context of Python for Finance & Quant Analysis.
- Apply NumPy to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

evaluation, NumPy, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a evaluation result produced with NumPy?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 09 / 18 MIN

## Reporting with Portfolio

### Learning objectives

- Explain reporting in the context of Python for Finance & Quant Analysis.
- Apply Portfolio to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

reporting, Portfolio, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a reporting result produced with Portfolio?

**Evidence to keep**

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## MODULE REVIEW

# Turn the module into usable evidence

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## MODULE 04

# Portfolio: Risk and governance

Apply Portfolio through security, regulation, review.

## LESSON 10 / 22 MIN

## Security with Quant

### Learning objectives

- Explain security in the context of Python for Finance & Quant Analysis.
- Apply Quant to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

security, Quant, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a security result produced with Quant?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 11 / 26 MIN

## Regulation with Python

### Learning objectives

- Explain regulation in the context of Python for Finance & Quant Analysis.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

regulation, Python, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a regulation result produced with Python?

### Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

## LESSON 12 / 30 MIN

## Review with yfinance

### Learning objectives

- Explain review in the context of Python for Finance & Quant Analysis.
- Apply yfinance to a bounded practical task.
- Evaluate the result using explicit quality criteria.

### Key terms

review, yfinance, finance

### Retrieval prompt

In Python for Finance & Quant Analysis, which evidence best supports a review result produced with yfinance?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

## MODULE REVIEW

# Turn the module into usable evidence

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Without checking the lessons, explain the module's three most important ideas in your own words.

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## CAPSTONE PROJECT

# Produce a reviewable Python for Finance & Quant Analysis project using Python, yfinance, NumPy.

|                    |                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------|
| <b>DELIVERABLE</b> | A working Python for Finance & Quant Analysis artefact plus an evidence-based self-review.       |
| <b>TOOLS</b>       | A suitable Python environment, A plain-text decision log, Test data or realistic sample material |

## Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Python.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

## Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

### RESPONSIBLE PRACTICE

This course is general education, not financial, investment, tax or legal advice.

## PROJECT EVIDENCE

# Document decisions another person can review

**1. PURPOSE AND USER**

State the intended user, need, expected outcome and constraints.

**2. INPUTS AND ASSUMPTIONS**

List source files, data, versions, permissions and assumptions.

## PROJECT EVIDENCE

# Tests, limitations and revision

**3. NORMAL CASE**

Expected result, observed result and evidence location.

**4. BOUNDARY OR FAILURE CASE**

What was difficult, what happened and why it matters.

**5. REVISION AND LIMITATION**

What changed, what improved and what remains unproven.

## REVIEW AND SOURCES

# Make the work credible and repeatable

## Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

**FINAL DECISION**

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

## Authoritative references

### National Payments System

Central Bank of Kenya - reviewed 2026-08-21

<https://www.centralbank.go.ke/national-payments-system/>

### Financial education

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/financial-education.html>

**SOURCE NOTE**

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

**NEXT IMPROVEMENT**

Choose one weakness found during review and improve it before extending the Python scope.